

AI3602 Data Mining Homework 1.3

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Question 1

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

For an arbitrarily fixed subset S , the probability of picking k elements from S is

$$P(|S \cap T| = k) = \frac{C_k^m C_{m-k}^{n-m}}{C_m^n},$$

and for each possible k , the jaccard distance is $\frac{k}{2m-k}$.

So the expected similarity is

$$E[J(S, T)] = \sum_{k=0}^m \frac{C_k^m C_{m-k}^{n-m}}{C_m^n} \cdot \frac{k}{2m-k}.$$

Original Questions

1. Suppose we have a universal set U of n elements, and we choose two subsets S and T at random, each with m of the n elements. What is the expected value of the Jaccard similarity of S and T ? A sum expression of the expectation is acceptable if you can't simplify it.

Figure 1: Question 1

Reference

None.